

Object Oriented Programming

Lab Assessment 1 (10 Marks)

Task-1 (Click [here](#) to open the compiler) (3 Marks)

Write a code that calculates the value of pi from the infinite series (1000 terms).

$$pi = 4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + \dots$$

Print a table that shows the approximate value of π after every 100th term of this series.

Task-2 (Click [here](#) to open the compiler) (3 Marks)

A palindrome is a number or a text phrase that reads the same backward as forward. For example, each of the following five-digit integers is a palindrome: 12321, 55555, 45554 and 11611. Write a program that reads in a five-digit integer and determines whether it's a palindrome.

Task-3 (Click [here](#) to open the compiler) (4 Marks)

Write a program that displays the following menu:

Geometry Calculator

1. Calculate the Area of a Circle
2. Calculate the Area of a Rectangle
3. Calculate the Area of a Triangle
4. Quit

Enter your choice (1-4):

If the user enters 1, the program should ask for the radius of the circle and then display its area. Use the following formula: πr^2 . Use 3.14159 for π and the radius of the circle for r .

If the user enters 2, the program should ask for the length and width of the rectangle and then display the rectangle's area. Use the following formula: $\text{area} = \text{length} * \text{width}$

If the user enters 3, the program should ask for the length of the triangle's base and its height, and then display its area. Use the following formula: $\text{area} = \text{base} * \text{height} * .5$

If the user enters 4, the program should end.

Input Validation: Display an error message if the user enters a number outside the range of 1 through 4 when selecting an item from the menu. Do not accept negative values for the circle's radius, the rectangle's length or width, or the triangle's base or height.